

Effects of different cooking methods on nutritional and physicochemical characteristics of selected vegetables.



The objective of the present study was to evaluate the effect of three common cooking practices (i.e., boiling, steaming, and frying) on phytochemical contents (i.e., polyphenols, carotenoids, glucosinolates, and ascorbic acid), total antioxidant capacities (TAC), as measured by three different analytical assays [Trolox equivalent antioxidant capacity (TEAC), total radical-trapping antioxidant parameter (TRAP), ferric reducing antioxidant power (FRAP)] and physicochemical parameters of three vegetables (carrots, courgettes, and broccoli). Water-cooking treatments better preserved the antioxidant compounds, particularly carotenoids, in all vegetables analyzed and ascorbic acid in carrots and courgettes. Steamed vegetables maintained a better texture quality than boiled ones, whereas boiled vegetables showed limited discoloration. Fried vegetables showed the lowest degree of softening, even though antioxidant compounds were less retained. An overall increase of TEAC, FRAP, and TRAP values was observed in all cooked vegetables, probably because of matrix softening and increased extractability of compounds, which could be partially converted into more antioxidant chemical species. Our findings defy the notion that processed vegetables offer lower nutritional quality and also suggest that for each vegetable a cooking method would be preferred to preserve the nutritional and physicochemical qualities.